Low-dose arachidonic acid intake increases erythrocytes and plasma arachidonic acid in young women.

Arachidonic acid (ARA) is considered to be a minor contributor to the diet. Previous reports regarding the effect of ARA supplementation on the composition of long-chain polyunsaturated fatty acids (LCPUFA) in the blood of humans are extremely limited. In the present study, we conducted a crossover double-blind, placebo-control study. Twenty-three young Japanese women consumed one capsule containing triacylglycerol enriched with 80 mg ARA, equivalent to the amount in one egg, daily for 3 weeks. Blood samples were drawn before and after treatment periods, and the compositions of the LCPUFA in blood lipid fractions were measured. The supplementation of ARA increased the composition of ARA, but did not decrease the composition of n-3LCPUFA in erythrocyte phospholipids and plasma phospholipids, esterified cholesterol, and triacylglycerol. We found that dietary ARA increased the ARA level in all lipid fractions of the blood, even at a very low dose.